# AI & MACHINE LEARNING – QUESTIONS

**Q.1. What is Machine Learning?**
**(8)**

**Q.2. Explain supervised & unsupervised learning.**
**(10)**

**Q.3. What is overfitting?**
**(9)**

**Q.4. Explain neural networks.**
**(5)**

**Q.5. What is gradient descent?**
**(9)**

**Q.6. Classification vs regression.**
**(8)**

**Q.7. Activation functions in DL.**
**(7)**

**Q.8. Bias-variance tradeoff.**
**(9)**

**Q.9. Confusion matrix use.**
**(8)**

**Q.10. Precision, Recall, F1-score.**
**(7)**

**Q.11. What is a CNN?**
**(6)**

**Q.12. What is an RNN?**
**(9)**

**Q.13. What is transfer learning?**
**(9)**

**Q.14. Explain ensemble methods.**
**(5)**

**Q.15. What are hyperparameters?**
**(8)**

**Q.16. Clustering algorithms.**
**(7)**

**Q.17. What is NLP?**
**(9)**

**Q.18. Importance of feature engineering.**
**(8)**

**Q.19. Train/validation/test split.**
**(8)**

**Q.20. What are Transformer models?**
**(7)**